Literature Review

You will Learn About

- ✓Purpose of Literature Review
- ✓Literature Resources and Techniques
- ✓Conducting a literature review
- ✓Referencing and Citation
- ✓Evaluating literature review
- ✓Identifying problem: Research question / problem statements and hypothesis
- ✓Case Study: Prepare a review paper.

Purpose of Literature Review

- Before Topic Chosen
  - Suitable research idea and relevant research materials
- After Topic Chosen
  - Gather and present evidence to support your claim that you have created some new knowledge
  - Assemble evidence support their claims that:
    - the topic is worthwhile;
    - The research does not merely repeat the work of others;
    - the researcher has created some new knowledge that was not known before.

Purpose and Scope of Literature Review

- Review of related literature
  - What has been done (searching, organizing, summarizing literature on a topic)
  - What needs to be done
- Several guidelines are important:
  - Do not include everything
  - Only review work that is related to your topic and research questions
  - For new topics, review literature that is related to your topic (Sheperis et al., 2017)

The Roles of the Literature

- Define the perimeters of the investigator's field.
- Placing the research question in perspective.
- Learning which methods and instruments have been useful and those that have less potential.
- Avoiding unintentional replication of previous studies
- Placing researchers in a better position to interpret the significance of their own results.

Steps for Writing your Literature Review

- Identify a topic for investigation
- Locate the relevant literature
- Critically evaluate the existing literature.
- Organize the quality and relevant information outlining the writing process.
- Present the information identifying the existing research and the purpose and benefits of the new research.

(Eaves, Sheperis, Craft, Frasier, and Wells, 2008)

Objectives of a literature review

- Show that the researcher is aware of existing work in the chosen topic area.
- Place the researcher's work in the context of what has already been published.
- Point to strengths, weaknesses, omissions or bias in the previous work.
- Identify key issues or crucial questions that are troubling the research community.
- Point to gaps that have not previously been identified or addressed by researchers.
- Identify theories that the researcher will test or explore by gathering data from the field.
- Suggest theories that might explain data the researcher has gathered from the field.
- Identify theories, genres, methods or algorithms that will be incorporated in the development of a computer application.
- Identify research methods or strategies that the researcher will use in the research.
- Enable subsequent researchers to understand the field and the researcher's work within that field.

Researching a Topic

Area of Interest

- Begin with an area of interest but avoid topics that are
  - too general
  - too specific
  - overused
  - outdated
  - do not warrant further investigation

Evaluating the Literature

- You must search and sift through all of the information encasing the topic of interest and ask:
- Are there loopholes in existing research?
- What are the benefits of existing studies and will a new study add to the literature?
- What are the limitations of the study under consideration?

(Sheperis, Young, and Daniels, 2017)

Contributions to the Existing Literature

- Conducting a literature review will orient the researcher with existing literature.
- The researcher will seek to demonstrate how a topic or question will make a contribution to existing literature.
- The researcher seeks to demonstrate that work is not an unnecessary replication of a study.

(Sheperis, Young, and Daniels, 2017)

Is the Topic Manageable?

Ask yourself:

- Am I truly interested in this topic?
- Does it meet all the assignment/personal requirements?
- Do I have access to enough information?
- Is a paper on this topic feasible?

(Sheperis, Young, and Daniels, 2017)

Literature Resources

- Books
- Manuals
- Journals
- Conference and Workshop Proceedings
- Reports
- Newspapers, magazines, radio and television
- Art exhibition catalogues
- Multimedia ‘literature’
- Resource catalogues and online databases
- Academic gateways (also called portals)
- People
- The Internet and Literature Reviews

Some highly rated journals in IS and computing

- *ACM Computing Surveys*
- *Communications of the ACM*
- *Digital Creativity*
- *IEEE Transactions*
- *Information Systems Research*
- *MIS Quarterly*

ACM DL DIGITAL LIBRARY

Google Scholar

IEEE Xplore DIGITAL LIBRARY

Literature Resources and Techniques

- Relevant literature may take several forms including **books, book chapters, published journal articles, conference papers or presentations, theses or doctoral dissertations**.

- When choosing sources consider:
  - Is it scholarly work?
  - Does it have application to the topic?
  - Is it based on theory?

(Sheperis, Young, and Daniels, 2017)

Peer Reviewed Journals

- Published journal articles are a common source of literature.
- Published journal articles require an editorial board to review the work, make corrections, and determine if it can be published.
- The work has been reviewed, edited, critiqued and corrected.

Is the Source Trustworthy?

- Researchers may want to consider the following questions when examining existing literature:
- Is the author an expert?
- Is the source current?
- Is the source complete?
- Is the source biased or unbiased?
- Is the source accurate and logical?

(VanderMey et al., 2011; Sheperis et al., 2017)

Other Sources

- Internet Sources (Be cautious)
- Primary Sources
  - Work that is considered the original report of research providing firsthand information or is the first published account is known as a primary source.
- Secondary Sources (Second hand accounts)
  - An assembled review of primary sources
  - May include textbooks, magazines, newspapers, television, radio, journal articles, documentaries, encyclopedias and nonfiction books.

(VanderMey et al., 2011; Sheperis et al., 2017)

Get Familiar with the Library

- Libraries
  - Libraries often hold archival pieces of information and other types of primary sources that cannot be retrieved online. Good researchers get acquainted with the library catalog, reference desk, and programs and systems for accessing information.
- Online databases
  - Online databases are electronic sources the library pays a fee to have access to. Published articles are found through electronic databases. They search thousands of journals, books, chapters in edited books, government documents, newspapers, and theses/dissertations.

(VanderMey et al., 2011; Sheperis et al., 2017)

Conducting a literature review

- Searching

Concept 1	Concept 2	Concept 3	Concept 4
Attitudes	Advertising	Mobile phones	Text messaging

Figure 6.1 Splitting a research topic into separate concepts

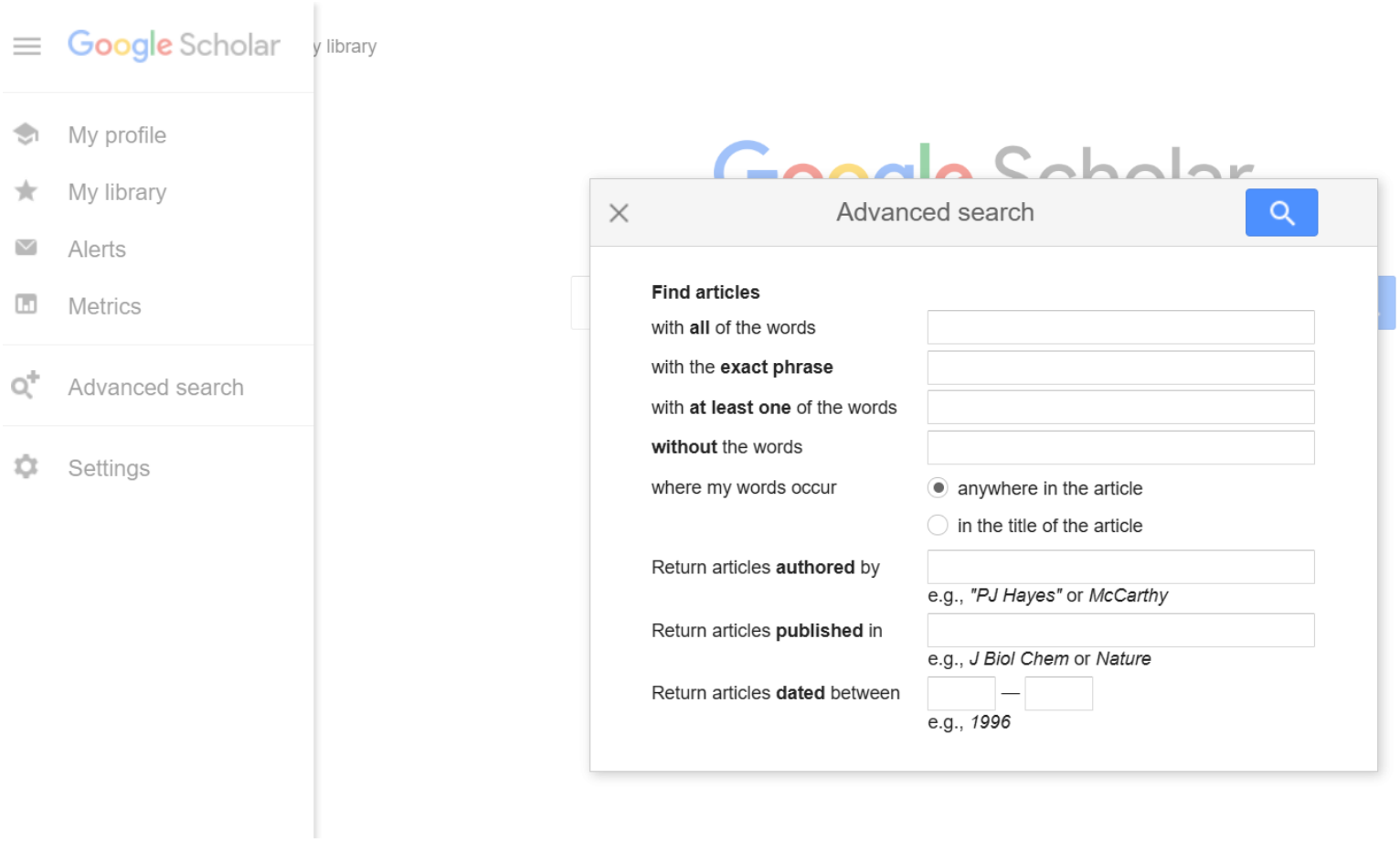
Concept 1	Concept 2	Concept 3	Concept 4
Attitudes	Advertising	Mobile phones	Text messaging
Beliefs	Marketing	Cell phones	SMS
Prejudices	Selling	Telephones	

Figure 6.2 Alternative terms for research concepts

Example	Outcome
attitud*	Results contain words such as attitude, attitudes, attitudinal
mobile AND phones	Results contain both terms, but they do not necessarily occur together
mobile OR phones	Results contain at least one of these terms
'mobile phones'	Results contain the phrase 'mobile phones'

Figure 6.3 Using symbols and Boolean operators in a search

Google Scholar Search



https://guides.library.ucsc.edu/c.php?g=745384&p=5361954

Conducting a literature review

- Obtaining
- Assessing
  - Questions that you could ask yourself about a book include:

    - Is the author someone eminent in the field?
    - Have you heard of the publisher before?
    - Is the publisher a university press (for example, Oxford University Press)? This might reassure you about the academic quality of the work.
    - Is the book in a second or subsequent edition, or has it been reprinted? You can find this out from the page that gives the publishing date, usually between the title and contents pages. If there has been more than one edition or reprint, this suggests that there has been a high demand for the book, implying others think it has something worthwhile to say.

  - For a journal article, you need to assess the journal itself. For example:

    - Is it aimed at academics or practitioners? You should concentrate on those aimed at academics.
    - How long has the journal existed? You can tell this by looking at the volume number: the higher it is, the longer the journal has existed. One that has existed for a long time is likely to have established itself as a respectable journal. However, the converse is

Conducting a literature review

not true. IS and computing are rapidly evolving fields, so new journals appear regularly. It is not valid to infer that a journal that has not existed for long cannot be a respectable source.

- Does the journal provide a list of its editorial board and advisors, and do these people seem to be of high standing? If they are, then the journal is probably an acceptable resource.
- Does the journal state clearly its policy for reviewing articles? Are all articles peer reviewed, some, or none at all? You should concentrate on those articles that are peer reviewed.

SJR

Scimago Journal & Country Rank



Conducting a literature review

- For conferences and workshops, you need to ask:
  - Is it an academic conference or one aimed solely at practitioners?
  - Is the conference/workshop well established? As with journals, conferences that have been held annually for many years are likely to be respected, but the converse is not necessarily true.
  - Does the conference/workshop give details of its programme committee? Do they appear to be scholars of high standing?
  - Does it make clear its reviewing policy, or were all papers accepted for presentation?
- Reading
- Critically evaluating
- Recording
  - a brief summary of the content – to act as an aid to your memory;
  - a brief summary of your evaluation – again to aid your memory;
  - bibliographic details of the literature source, so that you, your examiners and anyone else who reads your paper or thesis can find the text again (see the box on the facing page).

Conducting a literature review

- Writing a critical review

gathering and presenting evidence to support your claim that you have created some new knowledge, specifically that:

- the topic is worthwhile;
- the research does not merely repeat the work of others (or if it does, there is a deliberate reason for this);
- You have created some new knowledge that was not known before.

	Concepts			
Articles	A	B	C	...
Atkins (1996)	*	*		
Bell (2000)			*	
Jones (1995)	*			
Smith (1996)		*		
...				

Figure 6.4 Using a matrix to map papers to concepts

- Plagiarism

Conducting a literature review

Literature review: Making a reading grid

Source: https://bit.ly/2E7MF6k

Source information	Junco (2015)	Karpinski, Kirschner, Ozer, Mellot & Ochwo (2013)	Kirschner & Karpinski (2010)	Lau (2017)
Research topic/question	Relationship between time on various FB activities and GPA	Multitasking impact on relationship between FB & academic achievement	Are there differences in academic performance of student FB users and non-users?	Non-academic FB use bad for GPA. Academic FB use good for GPA. Multitasking negative.
Methodology	• Survey: quant > USA > 1399 undergrad sts	• Survey: quant & qual > USA & Europe > 590 undergrad sts > 285 grad sts	• Survey: quant & qual > USA > 102 undergrad sts > 117 postgrad sts	• Survey: quant > Hong Kong > 348 undergrad sts
Findings	• FB bad for academic performance • Poss. good for engagement	• Negative relation between FB use and GPA • Negative effects of multitasking	• FB users lower GPA • Similar online time for FB & non FB users • Multitasking negative	• Non-acad. FB use & multitasking negative for academic performance
Limitations	• Cross-sectional & correlational • Focus on GPA	• Correlational • Focus on GPA	• Correlational • Acad./learning use of FB not considered	• Cross-sectional • Multitasking activities unclear
Areas for future research	• Engagement impact & FB use with high acad. performance?	• Focus on how FB is used by high performing sts?	• Focus on how FB is used by high performing sts?	• More focus on multitasking & high acad. performance?

Adapted from Auckland University of Technology (Library)

@OpenAcademics

Steps for Writing a Lit Review

- Planning
- Reading and Research
- Analyzing
- Drafting
- Revising



Referencing and Citation

Plagiarism

IEEE Referencing

https://researchguides.njit.edu/ieee-citation/ieeereferencing

Identifying problem: Research question / problem statements and hypothesis



- ✓ When looking for **research gaps across publications** you may have noticed sentences like:

…has/have not been…
(studied/reported/elucidated)
…is required/needed…
…the key question is/remains…
…it is important to address…

- ✓ These expressions often indicate gaps; issues or topics related to the main question that still hasn't been subject to a scientific study. Therefore, it is important to take notice of them: who knows if one of these sentences is hiding your way to fame.



Author Services

https://scientific-publishing.webshop.elsevier.com/research-process/what-is-research-gap/

Guidelines for Creating a Literature Review

- Create an outline and process goals to guide work for the project.
- Process goals are objectives needed to accomplish a task.
- Break the assignment into manageable pieces.
- Work on one task at a time.

(VanderMey et al., 2011; Sheperis et al., 2017)

Guidelines for Creating a Literature Review

- The use of section headers creates clear and logical flow.
- Organize the existing research by points of relevance and elaborate on conclusions in the existing body of research.
- Organize by theoretical foundation and draw relationships between the current project and previous studies.

(VanderMey et al., 2011; Sheperis et al., 2017)

Guidelines for Creating a Literature Review

- Rules of Thumb:
  - There should be a source for every citation and a citation for every source.
  - If a source is in your reference list, it should be used somewhere in the body of your paper.

(VanderMey et al., 2011; Sheperis et al., 2017)

Structure

- Broad Statements about the topic, but not global
- Demonstrate why the topic is important
- Discuss what has been written about the topic so far
- Introduce your intention with the topic
- Report findings that illustrate relationships between the current project and previous studies
- Provide a research question, that when answered will provide a valuable step forward

(VanderMey et al., 2011; Sheperis et al., 2017)

Organization

Topic: Treating Anxiety with Narrative Therapy

What is the human, societal, and financial impact of anxiety?

What causes anxiety?

What are the current methods of treating anxiety?

How successful are current treatments?

What is Narrative Therapy?

How has it worked to treat anxiety in the past?

Why might it work better than other approaches in this study?

Research questions

Hypotheses

(VanderMey et al., 2011; Sheperis et al., 2017)

Organization

- Include findings from other studies
- Report research that supports and challenges the need for the study
- Do not cite content from the literature review from another article, rather locate the primary source
- Synthesize citations from multiple sources throughout the paragraphs whenever possible
- Pull the findings from other articles first, then organize and write manuscript
- Determine how many references are needed

(VanderMey et al., 2011; Sheperis et al., 2017)

https://scientific-publishing.webshop.elsevier.com/research-process/writing-a-good-review-article/

References

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